

Resource-Constrained Aerial Small-Object Detection with YOLO26s and Coordinate Attention

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https://github.com/ZHY9981/Drone_yolo_7types

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Abstract

Aerial small-object detection poses a unique challenge: targets smaller than 30 pixels must be detected in high-resolution drone footage, while consumer GPUs (8 GB VRAM) cannot accommodate the high-resolution detection heads that small objects require. We present a customized YOLO26s architecture that addresses three compounding bottlenecks—positional information loss, small-object gradient starvation, and VRAM overcommitment—through a combination of Coordinate Attention (CoordAtt), a P3+P4 dual-head design, WIoU v3 loss, and a reduced TaskAlignedAssigner pixel threshold. Across 20+ systematically ablated experiments on a curated 7-class aerial dataset, our best model (V16.0) achieves 74.00% mAP@0.5 and 52.11% mAP@0.5:0.95 with only 7.03M parameters, operating within an 8 GB VRAM budget at 4.0 ms inference latency. We also document four failed experiments and six key insights, including the counterintuitive finding that a dual-head architecture outperforms triple-head by 2.1% mAP under memory constraints.

1 Introduction

Aerial object detection from drone-captured imagery is a problem of growing practical importance, with applications ranging from traffic surveillance to search-and-rescue operations. However, it exposes a fundamental tension in modern object detectors: small targets require high-resolution feature maps, but high-resolution feature maps require GPU memory that consumer hardware does not have.

Consider a pedestrian viewed from 100 meters altitude. At 800-pixel input resolution, this target occupies approximately 20×15 pixels—less than 0.05% of the image area. A standard YOLO detector with P3/8, P4/16, and P5/32 downsampling strides places this target on the P3 feature map (100×100 spatial resolution), where each cell corresponds to an 8×8 pixel region in the original image. A 20-pixel target spans roughly 2.5 cells—barely enough for the detector to localize it. Adding a P2/4 head (200×200 resolution) would help, but the TaskAlignedAssigner’s alignment matrix at P2 resolution exceeds 8 GB VRAM at batch size 4.

This paper documents a systematic attempt to maximize aerial detection performance under a fixed hardware constraint (RTX 5060 Laptop GPU, 8 GB VRAM). Rather than scaling hardware, every design decision is evaluated by both accuracy and memory cost. The key contributions are:

1. A per-scale Coordinate Attention (CoordAtt) injection strategy that preserves fine-grained spatial cues for small targets without the VRAM cost of a full P2 head.
2. A P3+P4 dual-head architecture that outperforms both the standard P3+P4+P5 triple-head and the P2+P3+P4+P5 quad-head on 8 GB hardware.

3. A 20-version ablation study isolating the effect of each architectural modification, loss function, and data preprocessing decision with all failure cases documented.

2 Problem Analysis

Aerial small-object detection faces three compounding difficulties:

2.1 Extreme Scale Variation

In aerial imagery, a freight truck (occupying 300×100 pixels at 50 m altitude) and a pedestrian (20×15 pixels) coexist in the same image. The detector must simultaneously handle targets spanning two orders of magnitude in pixel area. Standard multi-scale detection heads (P3 through P5) are designed for COCO-scale objects, where the smallest targets (e.g., tennis rackets, bottles) are still ~ 50 pixels—double the size of an aerial pedestrian.

2.2 Memory Constraints on Consumer Hardware

The RTX 5060 Laptop GPU provides 8 GB of VRAM. At batch size 4 and image size 800×800 , the standard YOLO26s triple-head configuration consumes approximately 6.8 GB during training, leaving only 1.2 GB of headroom. Adding a P2 head (feature map $200 \times 200 \times 25,600$ cells) causes the TaskAlignedAssigner to allocate a $B \times N_{\text{anchors}} \times N_{\text{gt}}$ alignment matrix: at P2 resolution with 7 classes and up to 200 ground-truth instances per image, this matrix alone exceeds 2.5 GB—pushing total VRAM usage past the 8 GB limit and causing an OOM error.

2.3 Class Imbalance

The aerial_v9 dataset exhibits extreme long-tail distribution: person (73,770 instances) and car (69,840) dominate, while freight (1,306) and small-bus (1,117) each represent less than 2% of training instances. Standard equal-weight classification loss causes the model to optimize almost exclusively for the majority classes.

3 Methodology

3.1 Coordinate Attention (CoordAtt)

Channel attention mechanisms like Squeeze-and-Excitation (SE) improve feature representation by modeling inter-channel dependencies, but they discard all spatial information through global average pooling. For small aerial targets, *where* a feature is located is as important as *what* it represents.

CoordAtt [1] decomposes 2D global pooling into two 1D feature encoding operations along the horizontal and vertical directions:

$$\mathbf{z}_c^h(h) = \frac{1}{W} \sum_{0 \leq i < W} \mathbf{x}_c(h, i), \quad \mathbf{z}_c^w(w) = \frac{1}{H} \sum_{0 \leq j < H} \mathbf{x}_c(j, w) \quad (1)$$

The two 1D encodings are concatenated, passed through a shared 1×1 convolution for cross-channel interaction, then split into separate attention maps $\mathbf{a}^h$ and $\mathbf{a}^w$. The attended output is:

$$\mathbf{y}_c(i, j) = \mathbf{x}_c(i, j) \times \mathbf{a}_c^h(i) \times \mathbf{a}_c^w(j) \quad (2)$$

Placement strategy. We place CoordAtt per-scale at P3 and P4 neck positions (immediately before the Detect head), rather than at the backbone P5 layer. Our V16 ablation (Section 4) shows this is critical: a backbone P5 CoordAtt must propagate signals through two nearest-neighbor upsample steps to reach P3 detection features, losing spatial precision. Per-scale placement directly enhances the 120×120 (P3) and 60×60 (P4) feature maps.

3.2 P3+P4 Dual-Head Architecture

Removing the P5 detection head reduces parameter count by 32% (from 10.3M to 7.03M for YOLO26s) and saves approximately 40% of detection-head VRAM. This freed memory is reallocated to larger P3/P4 feature channels, improving small-object feature quality.

The counterintuitive result is that the dual-head configuration *improves* mAP by +2.1% over triple-head (V20 vs V16 in our ablations). This is not because P5 is useless—the P5 head detects large objects (buses, freight trucks) effectively—but because under the 8 GB memory constraint, the memory saved by removing P5 allows larger P3/P4 channels that improve the detection of the majority of targets (person, cycle, car, truck), which collectively account for over 80% of instances.

3.3 WIoU v3 Loss

Standard CIoU loss treats all bounding box regression errors equally. WIoU v3 [2] introduces an attention-based outlier suppression mechanism that adaptively weights regression gradients based on the quality of each prediction:

$$\mathcal{L}_{\text{WIoU v3}} = r \cdot \mathcal{L}_{\text{IoU}}, \quad r = \frac{\beta}{\delta \alpha^{\beta-\delta}} \quad (3)$$

where $\beta = \mathcal{L}_{\text{IoU}}^* / \sqrt{\mathcal{L}_{\text{IoU}}}$ measures outlier degree, and α, δ are hyperparameters controlling suppression strength. Small aerial targets (high β due to difficult localization) receive amplified gradients, while large targets (low β) are unaffected. This modification costs zero additional VRAM.

3.4 TAL Positive-Sample Threshold

The default TaskAlignedAssigner (TAL) [3] uses a stride-tensor minimum of 8 pixels for positive-sample assignment. At P3 resolution (stride 8), this means any ground-truth center more than 8 pixels from an anchor point is labeled negative. For a 20-pixel pedestrian, this excludes over 80% of valid P3 anchor points from receiving positive gradients.

We reduce the threshold to 4 pixels. This gives small targets approximately $4\times$ more positive anchors per instance, directly improving recall on person and cycle by +6–7% each.

3.5 Class-Weighted Loss

We apply per-class multiplicative weights to the classification loss:

$$\mathbf{w} = [2.0, 3.0, 1.8, 1.5, 1.0, 1.0, 1.0] \quad (4)$$

corresponding to [person, cycle, bus, small-bus, car, truck, freight]. Cycle receives the highest weight ($3.0\times$) because it is both rare (14K instances) and small. Freight and small-bus, while rare, are large and easy to detect, so they use base weight.

4 Experiments

4.1 Dataset

The aerial_v9 dataset is curated from three public sources: aerial.v1i (Roboflow, CC BY 4.0), Aerial Vehicle Detection (MIT), and aerial.v3i (MIT). The curation process involved merging the three sources, unifying class taxonomies (e.g., motorcycle + bicycle → cycle, van → car), and manually inspecting approximately 500 random samples to remove label noise and add missing annotations. The final dataset contains 8,075 training, 2,224 validation, and 738 test images across 7 classes.

4.2 Training Configuration

All experiments use a fixed hardware and training protocol:

Table 1: Training configuration (all versions).

GPU	RTX 5060 Laptop 8 GB
Batch size	4 (P2 experiments: 2)
Optimizer	SGD, lr ₀ =0.01, momentum=0.937, weight_decay=0.0005
Epochs	200 (V8–V16), 120 (V17–V20)
Image size	640–960 (V16 uses 800)
Augmentation	mosaic=0.5, scale=0.3, no mixup/copy-paste
Loss	WIoU v3 (box), VFL (cls), DFL

Mixup and copy-paste augmentation are deliberately disabled for aerial data—our early experiments (V1–V3) showed they degrade performance by 1.5–2.0%, likely because they create unrealistic object placements (e.g., a car floating on a rooftop) when applied to aerial imagery with dense, structured layouts.

4.3 Ablation Study

Table 2 presents the key milestones from the 20-version ablation study. Each version modifies exactly one variable from the preceding baseline.

Table 2: Key ablation results. Full table in project documentation.

Version	Key Change	mAP@0.5	mAP@0.5:0.95	Δ
V8.0	Baseline (clean data)	67.81%	44.31%	—
V8.1	+ WIoU v3 loss	69.59%	47.00%	+1.78%
V10.0	+ CoordAtt + aerial_v9	70.51%	49.33%	+0.92%
V12.0	imgsz 640→800	71.58%	50.27%	+1.07%
V14.0	P3+P4 dual-head, imgsz=960	73.71%	52.65%	+2.13%
V16.0	P3+P4 + per-scale CoordAtt, 800	74.00%	52.11%	+0.29%
V17.0	+ P4 RepNCSPeLan4	73.52%	51.27%	−0.48%
V18.0	+ ECA / ASFF	72.91%	50.84%	−1.09%
V19.0	P4 wide channel 256→384	73.44%	51.63%	−0.56%
V20.0	P3+P4+P5 triple-head (clean)	73.29%	50.41%	−0.71%

Table 3 shows the per-class breakdown of V16.0, our production model.

Table 3: Per-class performance — V16.0 (best model).

Class	AP50	AP50-95	Precision	Recall
person	60.2%	24.7%	77.7%	50.9%
cycle	43.0%	17.6%	67.0%	36.3%
bus	91.8%	74.5%	91.3%	86.3%
small-bus	99.2%	83.3%	92.3%	99.4%
car	67.2%	47.3%	87.2%	51.4%
truck	60.0%	41.0%	63.8%	53.6%
freight	96.6%	76.6%	89.8%	95.0%
Overall	74.0%	52.1%	81.3%	67.6%

4.4 Failed Experiments

Four experiments failed to improve performance, and we document them as honest records:

- **V4.0 – Freeze+unfreeze 2-stage training (46.82% mAP@0.5).** Standard YOLO transfer-learning protocol (freeze backbone, train head, unfreeze all) performs worse than training from scratch. Hypothesis: COCO pretrained features encode object shape priors (e.g., “person = upright rectangle”) that are orthogonal to the overhead view of aerial imagery, where pedestrians appear as small blobs.
- **V7.0 – VisDrone noisy labels (55.25% mAP@0.5, Recall 50%).** VisDrone2019 annotations contain significant label noise (motorcycles mislabeled as bicycles, inconsistent labeling across frames). The 12.56% gap between V7 and V8 demonstrates that data quality trumps architectural sophistication.
- **V8.2 – Transfer-learning relay (69.71% mAP@0.5).** Fine-tuning V8.1 weights on the expanded aerial_v9 dataset with the default learning rate caused catastrophic forgetting—all 7 classes degraded simultaneously. The learning rate reset on a larger dataset destroyed learned feature representations.
- **V13.0 – P2 high-resolution head (66.68% mAP@0.5, OOM).** While theoretically beneficial for small objects, the P2 head’s 25,600-cell feature map exceeds 8 GB VRAM with the TaskAlignedAssigner. The OOM occurs during assignment matrix allocation, before any forward pass.

5 Key Findings

1. **Data quality beats architecture.** Switching from noisy VisDrone to curated aerial_v9 yielded +12.56% mAP@0.5 (V7 → V8)—more than all four architectural modifications combined.
2. **Dual-head outperforms triple-head on 8 GB.** Removing P5 saves 32% parameters and improves mAP by +2.1%. The memory saved enables larger P3/P4 channels that benefit the most common target classes.
3. **WIoU v3 is the highest-ROI change.** +1.78% mAP@0.5 with zero VRAM cost and approximately 5 lines of code changed.
4. **CoordAtt is cheap but effective.** Negligible parameter increase (+0.2M) for +2.4% Precision gain.

5. **TAL threshold directly boosts small targets.** A simple $8 \rightarrow 4$ pixel threshold change improves person and cycle by +6–7% each.
6. **Mixup/copy-paste harm aerial detection.** These augmentations degrade performance by 1.5–2.0% on aerial data, likely due to unrealistic spatial configurations.

6 Limitations and Future Work

6.1 Car Detection Bottleneck

Car detection on V16.0 (67.3% AP50) lags person (60.2%) in absolute terms but represents the largest opportunity for improvement given car’s high instance count (69,840). Current hypotheses:

- **Scale ambiguity:** Cars viewed from 50–100 m altitude span a wide range of apparent sizes (50–200 pixels) that may confuse the P4-level detector. A P2 head would help differentiate small cars from large cars, but remains OOM-bound.
- **Occlusion patterns:** Cars in urban aerial scenes are frequently partially occluded by trees, buildings, or other vehicles. The current augmentation pipeline does not include occlusion simulation.

Ongoing work explores SimOTA center-prior assignment and SGLoss-style adaptive grid selection [4] to recover P2-head benefits without the OOM cost.

6.2 Rare Class Generalization

Freight and small-bus each represent <2% of training instances. While their detection performance is high on the current validation set (96.6% and 99.2% AP50 respectively), these metrics reflect the limited visual diversity of rare-class examples. Cross-dataset generalization remains untested.

6.3 Field-Deployment Constraints

The current pipeline achieves 4.0 ms inference on RTX 5060 (batch=1, FP16, 800×800), which is sufficient for real-time analysis on a consumer laptop in an outdoor setting. The motivating use case — running detection on a single laptop GPU without cloud or desktop hardware — is met by this configuration. TensorRT quantization is planned for further latency reduction on lower-power edge devices.

7 Conclusion

This project demonstrates that consumer-grade hardware (RTX 5060 8 GB) is sufficient for competitive aerial small-object detection when architectural decisions are co-designed with memory constraints. The key insight is not that any single modification is revolutionary, but that four targeted changes—CoordAtt, dual-head design, WIoU v3, and TAL threshold tuning—each address a specific bottleneck ($\sim 2\%$ improvement) and compound to a meaningful overall gain (+6.19% from V8.0 baseline). The 20-version ablation log, including documented failures, provides a reproducible baseline for future work on resource-constrained aerial detection.

References

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